



Zeolite-encapsulated Cu nanoparticles with enhanced performance for ethanol dehydrogenation

Lu Lin^{a,1}, Peng Cao^{a,b,1}, Jifeng Pang^{a,1}, Zhinuo Wang^{a,c}, Qike Jiang^d, Yang Su^a, Rui Chen^e, Zhijie Wu^{b,*}, Mingyuan Zheng^{a,d}, Wenhao Luo^{a,*}

^aCAS Key Laboratory of Science and Technology on Applied Catalysis, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China

^bState Key Laboratory of Heavy Oil Processing and the Key Laboratory of Catalysis of CNPC, China University of Petroleum, Beijing 102249, China

^cSchool of Materials Science and Engineering, Dalian Jiaotong University, Dalian 116023, China

^dDalian National Laboratory for Clean Energy, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, China

^eSchool of Materials Science and Engineering, Nankai University, Tianjin 300050, China

ARTICLE INFO

Article history:

Received 20 June 2022

Revised 11 July 2022

Accepted 13 July 2022

Available online 16 July 2022

Keywords:

Zeolites

Copper nanoparticles

Encapsulation

Ethanol

Dehydrogenation

ABSTRACT

The selective dehydrogenation of ethanol to acetaldehyde is one of the pivotal reactions for biomass valorization, wherein the well-defined and robust Cu-based solid catalysts are highly desired for potential practical application. Herein, ultra-small Cu metal nanoclusters (~1.8 nm), encapsulated inside Silicate-1 (S-1), have been successfully fabricated by an in-situ approach. The as-synthesized Cu@S-1 catalyst shows a near quantitative ethanol conversion, a good acetaldehyde selectivity up to ~90 %, and more importantly an excellent stability with no apparent deactivation even after a TOS of 110 h in the continuous dehydrogenation of ethanol to acetaldehyde at 523 K. Comprehensive and thorough characterization studies reveal that the spatial constraint environment of the zeolite could not only efficiently restrict the movements of Cu entities, which thus prevents the metal sintering during catalysis, but also induce a predominant amount of well-retained Cu⁺ species (up to ~70 %) that play an essential role for the enhanced performance.

© 2022 Elsevier Inc. All rights reserved.

1. Introduction

Selective ethanol dehydrogenation to acetaldehyde is both fundamentally and technologically important and thus has addressed great attention recently [1–5]. In particular, ethanol, which can be produced from the renewable biomass resource easily and abundantly [6–9], is one of the key building blocks for the sustainable production of many value-added chemicals and fuels, such as *n*-butanol, butadiene, ethyl acetate, peracetic acid, pentaerythritol and so on [10–15]. Note that many of these upgrading routes actually involve acetaldehyde as an intermediate, a key building block in its own right [5], endowing the dehydrogenation of ethanol to acetaldehyde with significant importance in biomass valorization.

Cu-based catalysts have shown to be efficient and promising with high activity and good selectivity to acetaldehyde in ethanol dehydrogenation [16–19]. However, the major challenge of employing Cu-based catalysts lies in their quick deactivation owing to the facile agglomeration of Cu species during catalysis,

as Cu possesses a low Tamman temperature and thus metal movements such as migration, coalescence, or Ostwald ripening can occur easily during catalysis [20–23]. In addition, Cu⁺ species have been reported to be the most active species in the ethanol dehydrogenation [24,25], and maintaining the relatively high proportion of Cu⁺ species in the Cu-based catalysts could play a pivotal role in attaining the superior catalytic performance for ethanol dehydrogenation. Furthermore, a neutral and suitable support, which can ensure a good dispersion of the Cu active species and efficiently prevent the further acid/base-catalyzed side-reactions of the active acetaldehyde [4,26–28], should be selected for efficient catalysis.

Zeolite-encapsulated metal nanoparticles, have attracted great attention for the preparation of ultrafine metal nanoparticles, which have shown significant advances in catalytic performance and great potential in biomass valorization [29–33]. The confined environment of zeolite cavities could not only provide the spatial restriction for the small metal entities, which can improve the metal stability against sintering of active metal species during catalysis, but also offer a microenvironment for tuning the metal properties. In our previous work, zeolite-encapsulated Cu nanoparticles catalysts could afford a relatively high proportion of Cu⁺

* Corresponding authors.

E-mail addresses: zhijiewu@cup.edu.cn (Z. Wu), w.luo@dicp.ac.cn (W. Luo).

¹ These authors contributed equally to this work.

species up to 71 %, indicating that the metal encapsulation inside the zeolite is an efficient approach for the preparation of the catalysts with the predominant Cu^+ species. The possible advantages of zeolite-encapsulated Cu nanoparticles have not yet been fully exploited for the selective ethanol dehydrogenation to acetaldehyde. The seldom examples available in the literature on the utilization of zeolitic Cu catalysts for the continuous conversion of ethanol to acetaldehyde at a high yield level during long period reactions (>100 h) have actually met with limited success. Notably, Li and coworkers recently prepared zeolite-confined Cu catalysts via a post-modulation approach, in which the zeolite H-beta was first fully dealuminated and metal species was subsequently impregnated into the zeolite cavities [34]. The as-obtained 5 % Cu/[Si]Beta afforded a good stability against Cu sintering with an acetaldehyde yield ~70 % for a time-on-stream (TOS) of 100 h in the ethanol-to-acetaldehyde conversion. Still, the acetaldehyde yield can be further improved by fine-tuning of the Cu-containing zeolites. Accordingly, the in-situ encapsulation of well-dispersed Cu nanoparticles during the assembly of purely siliceous zeolite can be an efficient and promising approach, which can meet all the key requirements for the high-performance Cu catalysts in the ethanol-to-acetaldehyde process.

Herein, well-dispersed and ultrafine Cu nanoparticles (~1.8 nm) encapsulated within the Silicate-1 zeolite (Cu@S-1) have been developed by an in-situ hydrothermal synthesis strategy. The continuous dehydrogenation of ethanol to acetaldehyde, which is one of key reactions with immense importance in biomass conversion, was performed over Cu@S-1 and two benchmark catalysts prepared via a post-impregnation method (Cu/S-1 and Cu/SiO₂) at 523 K. Compared to the quick and gradual deactivation shown for Cu/SiO₂ and Cu/S-1 catalysts after a TOS of < 20 h, the Cu@S-1 catalyst affords an excellent catalyst durability with a quantitative ethanol conversion and an acetaldehyde yield of ~90 % even after a TOS of 110 h in ethanol dehydrogenation. Comprehensive characterization studies by a combination of scanning transmission electron microscopy (STEM), N₂O chemisorption, Fourier transform infrared spectroscopy using CO as the probe molecule (CO-FT-IR), X-ray photoelectron spectra (XPS), X-ray-induced Auger electron spectroscopy (XAES) and temperature-programmed reduction of hydrogen experiments (H₂-TPR) were then performed to provide the insights into the enhanced catalyst performance of Cu@S-1 in the ethanol-to-acetaldehyde process.

2. Experimental

2.1. Preparation of materials

Synthesis of Cu@S-1: Tetraethyl orthosilicate (TEOS, 20.00 g, 99.7 %, Tianjin Kemiou) and tetrapropylammonium hydroxide (TPAOH, 11.40 g, 25 %, Tianjin Guangfu) were first added into deionized H₂O (27 mL) and stirred to form a clear solution. The obtained solution was then heated in a water bath and kept at 333 K for 1.5 h under stirring to form the starting gel of S-1. Then, Cu(NO₃)₂·3H₂O (0.43 g, 99 %, Tianjin Damao) was dissolved into deionized H₂O (10 mL) and mixed with tetraethylenepentamine (TEPA, 0.37 g, 95 %, TCI Shanghai), with the appearance of a dark blue solution. The obtained Cu-TEPA solution was added into the starting gel of S-1 and subjected at 443 K for 48 h under hydrothermal conditions in a Teflon-lined stainless-steel autoclave. The solid powder was obtained after separation by centrifugation, washing with deionized H₂O and drying at 373 K overnight. The obtained solid catalyst was calcined at 823 K for 6 h in air and subsequently cooled rapidly to room temperature. Cu@S-1 was then obtained and used after reduction at 673 K for 2 h under a flow of 5 vol% H₂/N₂.

Synthesis of Cu/S-1: TEOS (20.00 g) and TPAOH (11.40 g, 25 %) were first added into deionized H₂O (27 mL) and stirred to form a clear solution. The obtained solution was then heated in a water bath and kept at 333 K for 1.5 h under stirring to form the starting gel of S-1 and then subjected at 443 K for 48 h under hydrothermal conditions in a Teflon-lined stainless-steel autoclave. The solid powder was obtained after separation by centrifugation, washing with deionized H₂O and drying at 373 K overnight. The obtained solid catalyst was calcined at 823 K for 6 h in air to obtain S-1. For loading Cu species, Cu(NO₃)₂·3H₂O (0.076 g) was dissolved in deionized H₂O (1.45 mL), and then the aqueous solution was added into the S-1 (1.00 g) dropwise with stirring. After drying at room temperature for 12 h, the mixture was transferred to an oven at 353 K overnight, followed by calcination at 673 K for 4 h. Cu/S-1 was then obtained and used after reduction at 673 K for 2 h under a flow of 5 vol% H₂/N₂.

Synthesis of Cu/SiO₂: Cu(NO₃)₂·3H₂O (0.076 g) was dissolved in deionized H₂O (3.17 mL), and then the aqueous solution was added into the SiO₂ (1.00 g) dropwise with stirring. After drying at room temperature for 12 h, the mixture was transferred to an oven at 353 K overnight, followed by calcination at 673 K for 4 h. Cu/SiO₂ was then obtained and used after reduction at 673 K for 2 h under a flow of 5 vol% H₂/N₂.

2.2. Characterizations

X-ray diffraction (XRD) was carried out using the PW3040/60 X'Pert Pro powder X-ray diffractometer from PANalytical, the Netherlands. The diffraction source was Cu K α ray (λ = 1.5432 Å), Ni filtered, LynxExe array detector, tube voltage and tube current were 40 KV and 40 mA, respectively. The scanning step is 0.02°. The crystalline phase was identified with reference to the ICDD PDF-2 database. The actual Cu loading of catalysts was determined by inductively coupled plasma optical emission spectroscopy (ICP-OES) on an IRIS Intrepid II XSP instrument (Thermo Electron Corporation) with a sensitivity line of 349.8 nm and a detection range of 1–10 ppm. N₂ physisorption isotherms were obtained at 77 K using a Micromeritics ASAP 2460 instrument, all samples were outgassed at 473 K for 12 h prior to the physisorption measurements. Scanning transmission electron microscopy (STEM) images were obtained with a JEOL JEM-2100F microscope operated at 200 kV. Energy dispersive X-ray spectra (EDX-mapping) were collected with a JEOL Oxford X-MaxN 80 T silicon drift detector, by rastering the beam over the target zone. Aberration-corrected scanning transmission electron microscopy (AC-STEM) images in the high angle annular dark field mode (AC-HAADF-STEM) were acquired by using a JEOL JEM ARM-200F STEM operating at 200 kV, located in the electron Physical Sciences Imaging Centre at Dalian Institute of Chemical Physics. The copper dispersion and the surface area of metallic copper of catalysts were determined by N₂O chemisorption using Micromeritics Auto Chem II 2920 equipped with the thermal conductivity detector (TCD). Catalysts were first reduced in the procedure described in the temperature-programmed reduction (TPR) experiment in 10 % H₂/Ar at a flow rate of 30 mL/min until 673 K. The amount of hydrogen consumption in the first TPR was denoted as A₁. Then the reactor was purged with Ar at 323 K. A flow of 10 % N₂O/Ar (30 mL/min) was used to oxidize surface copper atoms to Cu₂O at 323 K for 1 h. The reactor was flushed with Ar to remove the oxidant. Finally, another TPR experiment was performed in 10 % H₂/Ar at a flow rate of 30 mL/min until 673 K. Hydrogen consumption in the second TPR was denoted as A₂. The dispersion (D_{Cu}) and the area of surface Cu (S_{Cu}) were calculated according to equations [35]:

$$D_{\text{Cu}} = \frac{2 \times A_2}{A_1} \times 100\%$$

$$S_{Cu} = \frac{2 \times A_2 \times N_{av}}{A_1 \times M_{Cu} \times 1.4 \times 10^{19}} (m_{Cu}^2/g_{Cu})$$

In these equations, N_{av} is Avogadro's constant, M_{Cu} is the relative atomic mass of copper (63.46 g/mol), and 1.4×10^{19} is the number of copper atom of per square meter, because the average surface area of copper atom is assigned as $7.11 \times 10^{-2} \text{ nm}^2$.

Average volume-surface diameter (d_{Cu}) can be expressed as a function [35]:

$$d_{Cu} = \frac{6000}{S_{Cu} \times \rho_{Cu}} (nm)$$

ρ_{Cu} in this equation is the density of copper (8.92 g/cm³).

Fourier transform infrared spectroscopy (FT-IR) was recorded on a BRUKER Equinox 55 spectrometer (MCT detector). Each spectrum contained 32 scans and was collected with a resolution of 4 cm^{-1} . For the spectra using CO as the probe molecule (FT-IR-CO), self-supported catalyst wafers were tableted at 3 kbar pressure for 5 s and then put inside a self-designed cell with a CaF_2 window. The cell was first vacuumed to 10^{-6} mbar and the wafer was then degassed at 573 K for 1 h with a heating ramp of 5 K/min. After cooling down to 83 K with liquid nitrogen, the cell was exposed to a CO gas chamber, which could precisely control the CO injection amount by monitoring the gas pressure in the cell. CO adsorption was studied at 83 K and stepwise increasing CO pressure, and CO desorption was also studied at 83 K during evacuation process, and followed by temperature-programmed desorption step under high vacuum conditions. X-ray photoelectron spectroscopy (XPS) and X-ray-induced Auger electron spectroscopy (XAES) experiments were conducted on a Thermo Fisher ESCALAB 250Xi, and spectra were collected using an aluminum anode (Al K α = 1486.6 eV). All obtained XPS spectra were calibrated by the adventitious carbon C 1 s band (284.6 eV), for correcting the energy shift caused by electrostatic charging. Temperature-programmed reduction of hydrogen experiments (H_2 -TPR) were performed on a Micromeritics Auto Chem II 2920 equipped with the TCD detector. Before the H_2 -TPR measurement, 0.1 g of the samples was treated at 673 K under Ar flow (30 mL/min) for 1 h to remove adsorbed species. After cooling to room temperature, the flowing gas was changed to a 10 % H_2 /Ar flow (30 mL/min), and the sample was heated to 1173 K at a ramping rate of 5 K/min.

2.3. Catalytic tests

Ethanol dehydrogenation was performed in a fixed-bed reactor with an inner diameter of 5 mm. Typically, a given amount of catalyst (10–40 mesh, 0.2 g) was loaded into the microtubular reactor.

Prior to each reaction, the catalyst was reduced in a H_2 flow at 573 K for 1 h. After the reactor was cooled down to reaction temperature, the gas was changed to nitrogen and ethanol was introduced to the reactor using a cooled bubbler (278 K). After passing through a hot trap (373 K), gaseous products were analyzed online using a gas chromatograph (GC) equipped with an auto-sampling valve, HP-FFAP column (30 m, 0.320 mm) and flame ionization detector (FID). Ethanol conversion and acetaldehyde yield was calculated based on the carbon mole ratio, using the following equations:

$$\text{Conversion (\%)} = \frac{\text{mol of ethanol fed into the reactor} - \text{mol of the residual ethanol}}{\text{mol of ethanol fed into the reactor}} \times 100\%$$

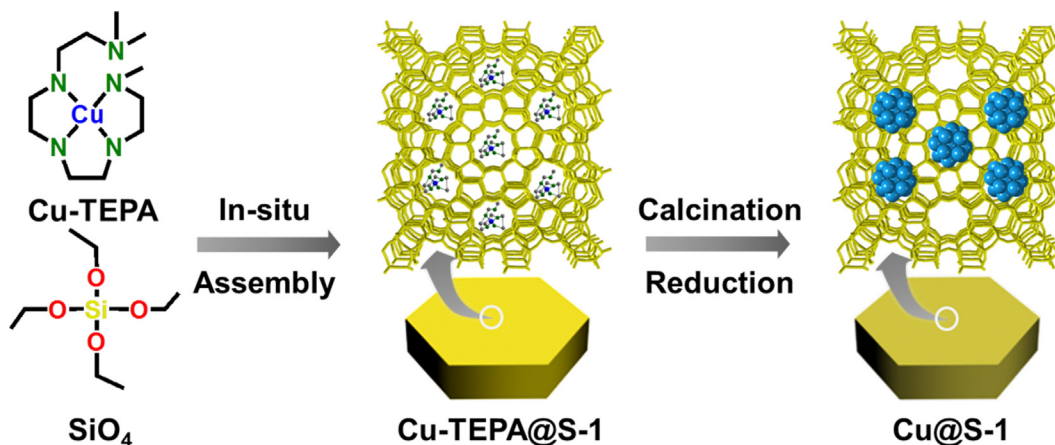
$$\text{Yield (\%)} = \frac{\text{mol of acetaldehyde in the product}}{\text{mol of ethanol fed into the reactor}} \times 100\%$$

3. Results and discussion

3.1. Physicochemical properties of different catalysts

Scheme 1 illustrates the in-situ synthesis strategy of Cu@S-1. In detail, Cu^{2+} cations first coordinate with polyethylenepolyamine (TEPA) to avoid precipitation under alkaline hydrothermal conditions during the zeolite S-1 synthesis. The as-obtained Cu-TEPA complexes could be captured during the zeolite assembly process and thus encapsulated within the cavities of the zeolite via the interaction between the metal complex and the zeolite [33,36]. After a subsequent calcination at 550 °C for 6 h in air and a following reduction at 400 °C for 2 h under 5 % H_2/N_2 atmosphere, well-dispersed Cu nanoparticles within the cavities of S-1 could be obtained, denoted as Cu@S-1. Cu/S-1 and Cu/SiO₂ as benchmark catalysts were also synthesized via a wet impregnation method.

X-ray diffraction (XRD) patterns of Cu/S-1 and Cu@S-1 show the characteristic diffraction peaks of an MFI topological structure (**Fig. S1a**), indicating that the addition of Cu species has limited impact on the formation of the S-1 zeolite. Cu/SiO₂ also retain the patterns of the supports after Cu loading (**Fig. S1b**). The absence of characteristic peaks of Cu species in all of the three Cu-containing catalysts is possibly due to the low Cu loading of ~2.00 wt%, as determined by inductively coupled plasma atomic emission spectroscopy (ICP-OES). **Table 1** lists the physicochemical properties of different samples. Compared with the parent S-1, Cu@S-1 shows a more apparent decrease in microporosity in terms of micropore surface area and micropore volume than Cu/S-1 after



Scheme 1. Schematic illustration of the in-situ encapsulation approach for the synthesis of Cu@S-1.

Table 1

Physicochemical properties of different zeolite catalysts.

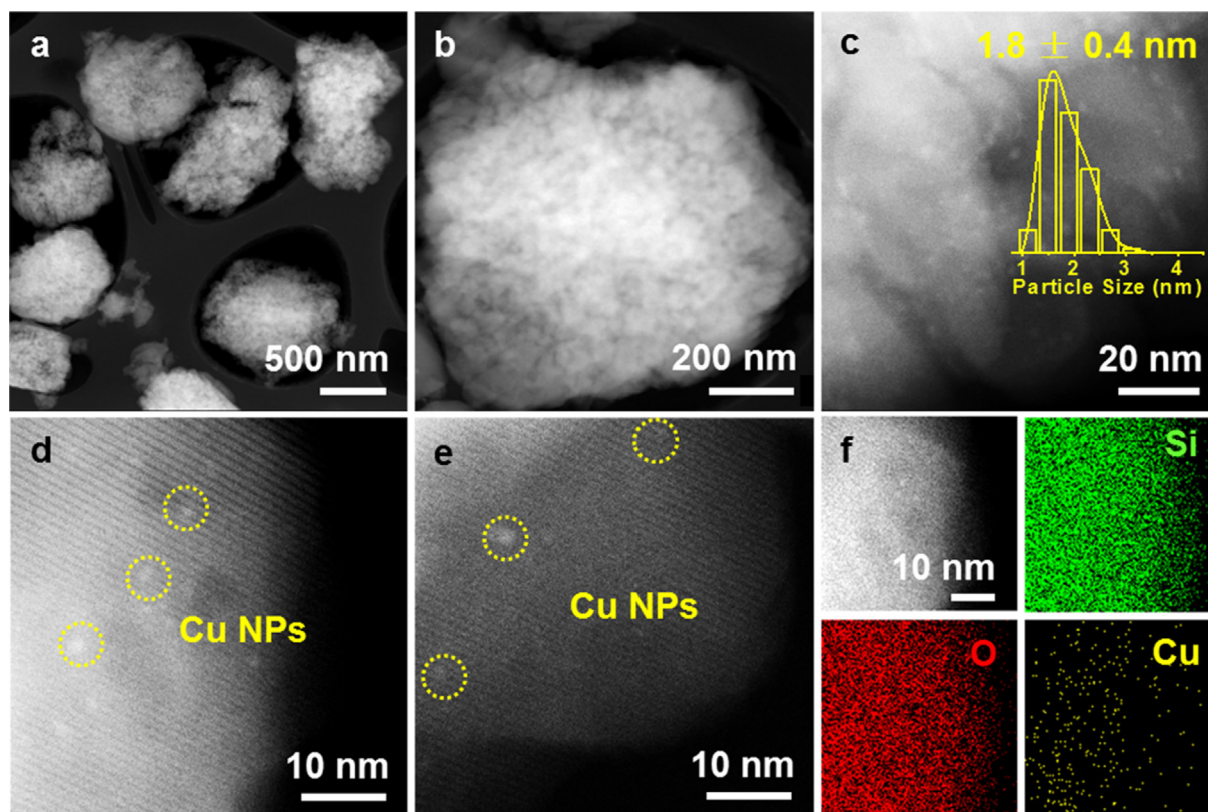
Samples	Cu loading ^b (wt%)	S_{BET} (m ² /g)	S_{Micro} ^c (m ² /g)	V_{Micro} ^c (cm ³ /g)	D_{Cu} ^d (%)	S_{Cu} ^d (m ² /g)	d_{Cu} ^d (nm)
S-1	–	413	214	0.112	–	–	–
Fresh Cu/SiO ₂	2.02	414	34	0.015	2.0	13.6	49.3
Fresh Cu/S-1	2.07	355	208	0.110	35.6	241.4	2.8
Fresh Cu@S-1	2.05	374	176	0.092	51.4	348.3	1.9
Spent Cu/SiO ₂ ^a	2.02	238	18	0.008	1.4	9.4	71.8
Spent Cu/S-1 ^a	2.02	308	199	0.104	27.0	183.2	3.7
Spent Cu@S-1 ^a	2.01	333	167	0.087	50.0	338.8	2.0

^a The spent Cu/SiO₂, Cu/S-1 and Cu@S-1 were obtained from the continuous dehydrogenation of ethanol to acetaldehyde after a TOS of 16 h, 18 h and 110 h, respectively. Reaction conditions: T = 523 K, ethanol gas/N₂ 40 mL/min, WHSV = 0.53 h^{−1}.

^b Obtained from ICP-OES analysis.

^c Micropore surface area (S_{Micro}) and Micropore volume (V_{Micro}) are determined by the *t*-plot method.

^d Cu dispersion (D_{Cu}), Cu surface area (S_{Cu}) and Cu average volume-surface diameter (d_{Cu}) are obtained from the N₂O chemisorption experiments.

**Fig. 1.** (a–c) HAADF-STEM images, (d–e) AC-HAADF-STEM images and (f) EDX mapping analysis of Cu@S-1.

Cu addition (Fig. S2 and Table 1), indicating the successful encapsulation of Cu species inside the micropores of the S-1 zeolite. In contrast, Cu/S-1 shows a major decrease in the external surface area and yet a limited loss of the microporosity after Cu addition compared to the parent S-1, which could point to the location of Cu particles majorly on the external surface of S-1.

The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of Cu@S-1 are shown in Fig. 1a–b. It is evident that hardly any large Cu aggregates are visualized in Cu@S-1. Analysis of the representative high-resolution images (Fig. 1c) reveals the formation of well-dispersed Cu nanoparticles with an average diameter of ~1.8 nm, almost identical with the average particle size estimated from N₂O chemisorption (Table 1, 1.9 nm). Notably, aberration-corrected HAADF-STEM (Fig. 1d–e) and the corresponding energy-dispersive X-ray (EDX) mapping analysis (Fig. 1f) further indicate that most of these

Cu nanoparticles are encapsulated within the S-1 support, in consistency with the results of N₂ physisorption. As for Cu/S-1 catalysts, although some large Cu particles of ~30 nm (circled in Fig. S3a–b) are clearly visualized on the external surface of the zeolite S-1, most Cu nanoparticles are with an average particle size of ~2.3 nm for Cu/S-1 (Fig. S3c), which is slightly smaller than the average particle size of 2.8 nm detected by N₂O chemisorption (Table 1). The EDX mapping results of Cu/S-1 confirm that the partial Cu species are likely located and agglomerated on the external surface of S-1 (Fig. S3d). Comparatively, the Cu/SiO₂ catalyst possesses the largest Cu particle size of ~48 nm, and the Cu particles are majorly located on the surface of SiO₂ (Fig. S4 and Table 1). Notably, the Cu@S-1 possesses a much higher Cu surface area (348.3 m²/g) than Cu/S-1 (241.4 m²/g) and Cu/SiO₂ (13.6 m²/g), indicating Cu@S-1 with a significant amount of well-dispersed and small Cu entities. This is consistent with previous observation

that the in-situ encapsulation approach could significantly improve the Cu dispersion and simultaneously attain small nanoparticles [33]. Obviously, the in-situ encapsulation strategy shows significant advantages in attaining the ultra-small and highly dispersed Cu nanoparticles compared to the traditional wet impregnation method.

3.2. Catalytic performance for ethanol dehydrogenation

The catalytic performances of Cu/SiO₂, Cu/S-1 and Cu@S-1 in the dehydrogenation of ethanol to acetaldehyde were evaluated in a fixed-bed reactor at 523 K. The feedstock is ethanol gas in N₂ atmosphere (40 mL/min) with a weight hour space velocity (WHSV) of 0.53 h⁻¹. Ethylene and ethyl acetate are detected as by-products (Fig. S5), which is separately derived from the dehydration of ethanol and the coupling between ethanol and acetaldehyde (Fig. 2a) [5]. Fig. 2b-c illustrates the time profiles of ethanol conversion and acetaldehyde yield over the three Cu-based catalysts. Cu/SiO₂ shows a quick deactivation, with a significant decrease in the ethanol conversion from 91.7 to 45.2 % and in the acetaldehyde yield from 79.5 to 32.1 %, respectively, after a time-on-stream (TOS) of 16 h. Similar deactivation of Cu/SiO₂ has also been reported by Ji and co-workers [24], in which the Cu/SiO₂ prepared via a wet impregnation method showed a ~50 % drop in ethanol conversion after a TOS of 25 h under similar conditions. For Cu/S-1, a gradual drop of the ethanol conversion from 97.7 to 86.8 % was observed after a TOS of 18 h. While the acetaldehyde yield increases from 82.1 to 88.2 % in the initial 2 h and then decreases to ~80.7 % after the following 16 h. Compared to Cu/SiO₂ and Cu/S-1, the Cu@S-1 catalyst prepared by the in-situ encapsulation strategy outperforms in both activity and stability. Cu@S-1 shows an ethanol conversion of 98.8 % at the initial stage

of the reaction, and even after a TOS of 110 h, the ethanol conversion can still retain at the level of ~98 % (97.8 %), indicating no catalyst deactivation. While the acetaldehyde yield increases from 73.7 to 80.6 % after a TOS of 23 h and is stabilized at ~90 % after a TOS of 71 h subsequently. Notably, the quantitative ethanol conversion and an acetaldehyde yield of 89.8 % can be sustained even after a TOS of 110 h, indicating an unprecedented stability of the high-performing Cu@S-1. Therefore, the Cu@S-1 prepared by the in-situ encapsulation strategy could enhance the catalytic performance in terms of activity and stability in the dehydrogenation of ethanol to acetaldehyde, compared to the Cu/S-1 and Cu/SiO₂ prepared by the wet impregnation method.

After the continuous dehydrogenation reaction, limited leaching of Cu species is observed over the spent Cu/SiO₂, Cu/S-1 and Cu@S-1 after a TOS of 16 h, 18 h and 110 h, respectively, as detected by ICP-OES (Table 1). XRD results of the spent Cu/S-1 and Cu@S-1 reveal that both catalysts retain the MFI zeolite structure upon catalysis (Fig. S1). The spent Cu/SiO₂ shows a severe loss of porosity with a drop in S_{BET} from 397 to 287 m²/g and a significant decrease in V_{micro} from 0.015 to 0.008 m³/g, possibly pointing to the severe coke deposition. This is also in line with the fact that quite an amount of ethylene (~10 %), which is likely to generate coke species deposited on the catalyst surface and further decrease the porosity of the catalyst [37–39], formed on Cu/SiO₂. Additionally, an obvious increase in the average Cu particle size from 49.3 to 71.8 nm is detected for the Cu/SiO₂ after a TOS of 16 h, as evidenced by STEM (Fig. S6) and N₂O chemisorption (Table 1), indicating a severe sintering of Cu particles upon catalysis. These results suggest that the quick deactivation of Cu/SiO₂ is attributed to the combined effects of coke deposition and the severe Cu sintering. By comparing the fresh and spent catalysts, limited change in porosity is observed for both Cu/S-1 and Cu@S-1 catalysts,

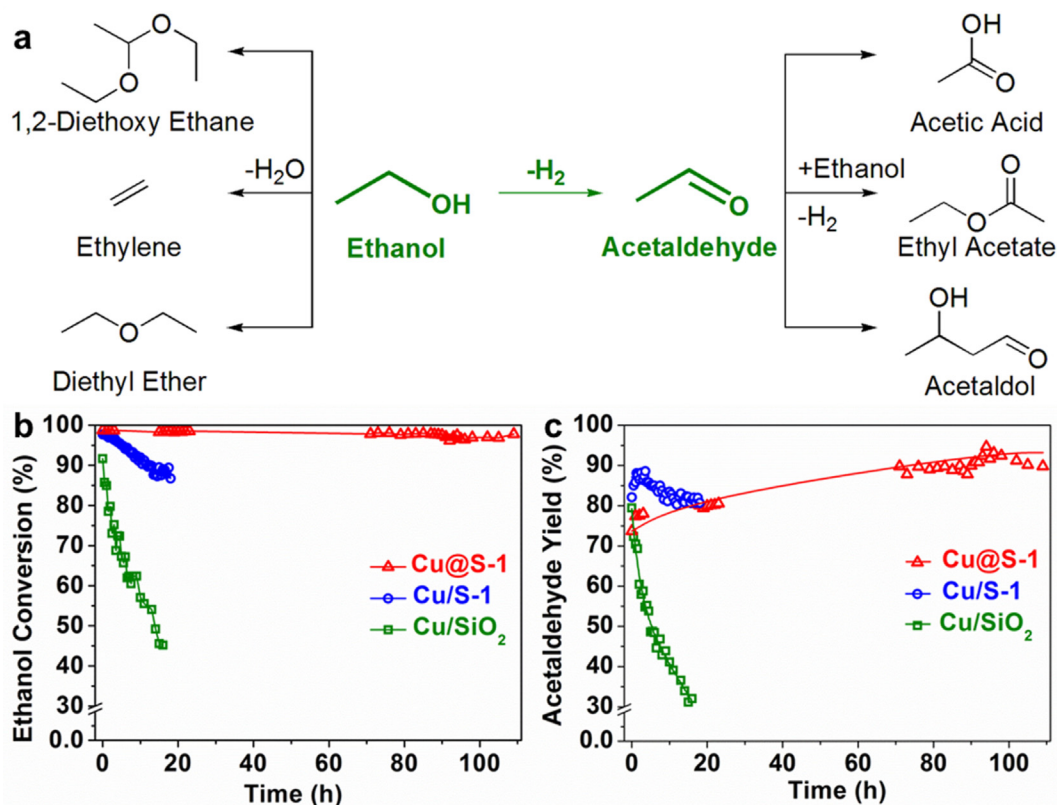


Fig. 2. (a) Typical reaction networks for ethanol dehydrogenation; (b-c) The time profiles of the catalytic dehydrogenation of ethanol to acetaldehyde over Cu/SiO₂, Cu/S-1 and Cu@S-1. Reaction conditions: T = 523 K, ethanol gas/N₂ 40 mL/min, WHSV = 0.53 h⁻¹.

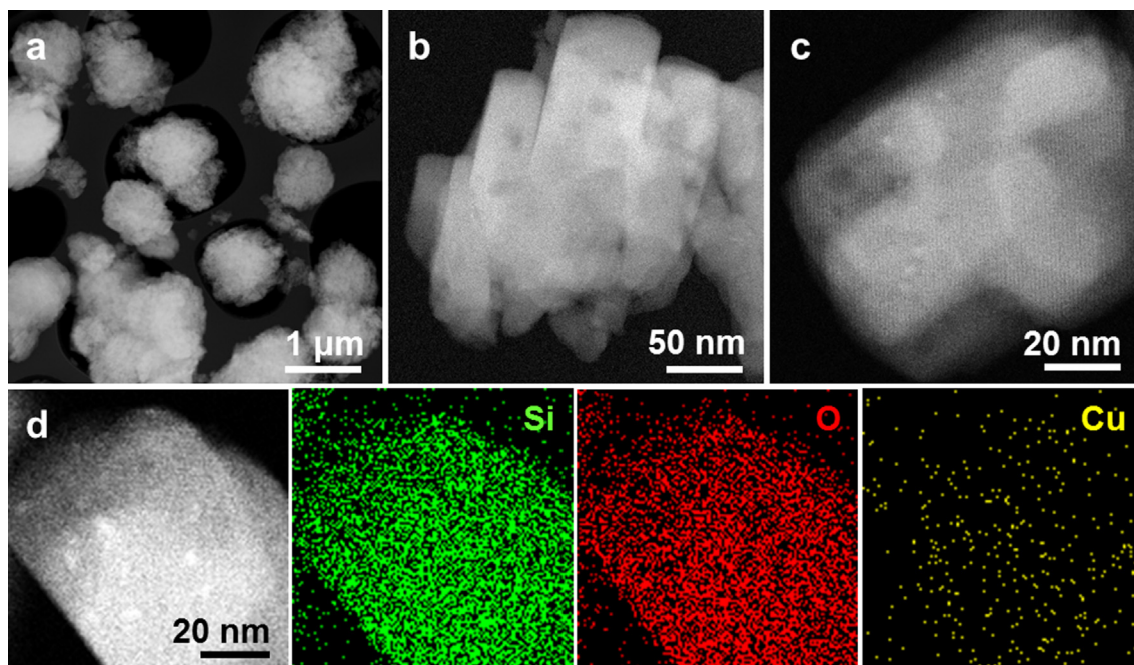


Fig. 3. (a–c) HAADF-STEM images and (d) EDX analysis of the spent Cu@S-1 catalyst after a TOS of 110 h. Reaction conditions: $T = 523$ K, ethanol gas/ N_2 40 mL/min, WHSV = 0.53 h^{-1} .

suggesting that the issue of coke deposition can be alleviated for employing the zeolite S-1 as support. However, similar phenomenon of Cu sintering is still observed for Cu/S-1, with an increase in the average Cu particle size from 2.8 to 3.7 nm after a TOS of 18 h (Table 1). Some large Cu aggregates with the size in the range of 20–50 nm are also visualized on the external surface of the spent Cu/S-1, as shown in the STEM images (Fig. S7). In contrast, limited changes in the average Cu particle size from 1.9 to 2.0 nm, as determined by N_2O chemisorption (Table 1), are observed for the Cu@S-1 after a TOS of 110 h. Additionally, no large particles are observed for the spent Cu@S-1 by HAADF-STEM (Fig. 3a–b) and small Cu nanoparticles are still highly dispersed inside the zeolite S-1, confirmed by HAADF-STEM and EDX-mapping analysis (Fig. 3c–d). Accordingly, the encapsulation of Cu species within the zeolite S-1 could efficiently inhibit the sintering of Cu particles, which is the major reason accounted for the deactivation of Cu-based catalysts (Cu/SiO₂ and Cu/S-1), thus leading to the sustained high-performance during the continuous dehydrogenation of ethanol to acetaldehyde.

3.3. Insights into catalytic performance

The Fourier transform infrared spectroscopy measurements using CO as the probe molecule (FT-IR-CO) were conducted for Cu/SiO₂, Cu/S-1 and Cu@S-1 catalysts at a low temperature of 83 K (Fig. 4). Due to the instability of Cu^{2+} -CO and Cu^0 -CO species, which renders this characterization associated with a number of challenges, FT-IR-CO is thus suitable for determining the state of Cu^+ species in the samples [40]. Cu/SiO₂ mainly displays one signal at 2139 cm^{-1} , referring to Cu^+ -CO species [41,42]. Comparatively, the spectra of Cu/S-1 and Cu@S-1 both show a slight red-shift of Cu^+ -CO signal to 2133 cm^{-1} . Such a shift might suggest that the environment of Cu^+ species in Cu/SiO₂ is with more positive charge species (i.e. Cu^{2+} species) than those in the Cu/S-1 and Cu@S-1. Notably, three additional signals at 2206, 2171 and 2147 cm^{-1} corresponding to Cu^+ species are observed for Cu@S-1, indicating the prevailing of Cu^+ species in Cu@S-1. The two features at 2171 and

2147 cm^{-1} can be attributed to the symmetric and antisymmetric vibrations of $Cu^+-(CO)_2$ species, respectively [41]. The weak signal develops at 2206 cm^{-1} , can be ascribed to $Cu^+-(CO)_n$ species [43,44], which is often detected for the samples with zeolite-encapsulated Cu^+ species [33].

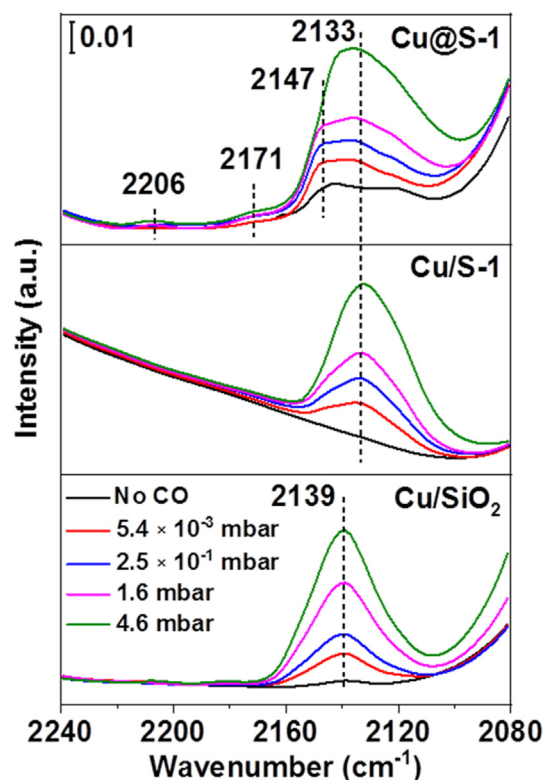


Fig. 4. FT-IR-CO spectra of Cu/SiO₂, Cu/S-1 and Cu@S-1, with stepwise adsorption of CO from 0 to 4.6 mbar at 83 K.

To further examine the chemical state of Cu species, X-ray photoelectron spectroscopy (XPS) were then employed for the three Cu-based catalysts. The surface Cu contents of the three different catalysts are measured and compared by XPS (Table S1). Note that the three catalysts bear almost a same content of ~ 2.00 wt% in the bulk phase. However, Cu@S-1 shows a much lower surface Cu content of 0.4 wt% compared to Cu/S-1 and Cu/SiO₂, again confirming that Cu nanoparticles should be dominantly encapsulated inside the zeolite S-1, rather than on the external surface. This is in good consistency with the results of N₂ physisorption and STEM, reflecting a successful metal encapsulation by the in-situ synthesis strategy. Fig. 5a shows the Cu 2p XPS spectra of the three Cu-based catalysts, and different characteristic Cu features can be discerned and quantified by peak deconvolution. For Cu/S-1 and Cu@S-1, the Cu 2p_{1/2} and 2p_{3/2} signals at 933.4 and 953.2 eV, respectively, are assigned to reduced Cu species (Cu⁰ and Cu⁺). The Cu 2p_{1/2} and

2p_{3/2} signals at 935.4 and 955.4 eV, respectively, are assigned to Cu²⁺ species [45–47]. While the residual two signals situated at 942.9 and 962.9 eV refer to the satellite peaks of Cu²⁺ species [47]. Compared with the signals for Cu/S-1 and Cu@S-1, the featured signals for Cu/SiO₂ show a shift of ~ 0.8 eV to the higher binding energies (reduced Cu species at 934.2 and 954.0 eV, Cu²⁺ species at 936.2 and 956.2 eV and satellite at 943.7 and 963.7 eV), in consistency with the results of FT-IR-CO that Cu⁺-CO species locate at higher wavenumber. This further indicates the more positively charged environment of Cu species in Cu/SiO₂ as compared to that in Cu/S-1 and Cu@S-1.

For further discrimination and quantification of Cu⁺ species from Cu⁰ species, Cu LMM X-ray-induced Auger electron spectroscopy (XAES) measurements were further conducted on the three Cu-based catalysts (Fig. 5b). The signals at 916.5 and 919.5 eV are separately referred to Cu⁺ and Cu⁰ species [48,49].

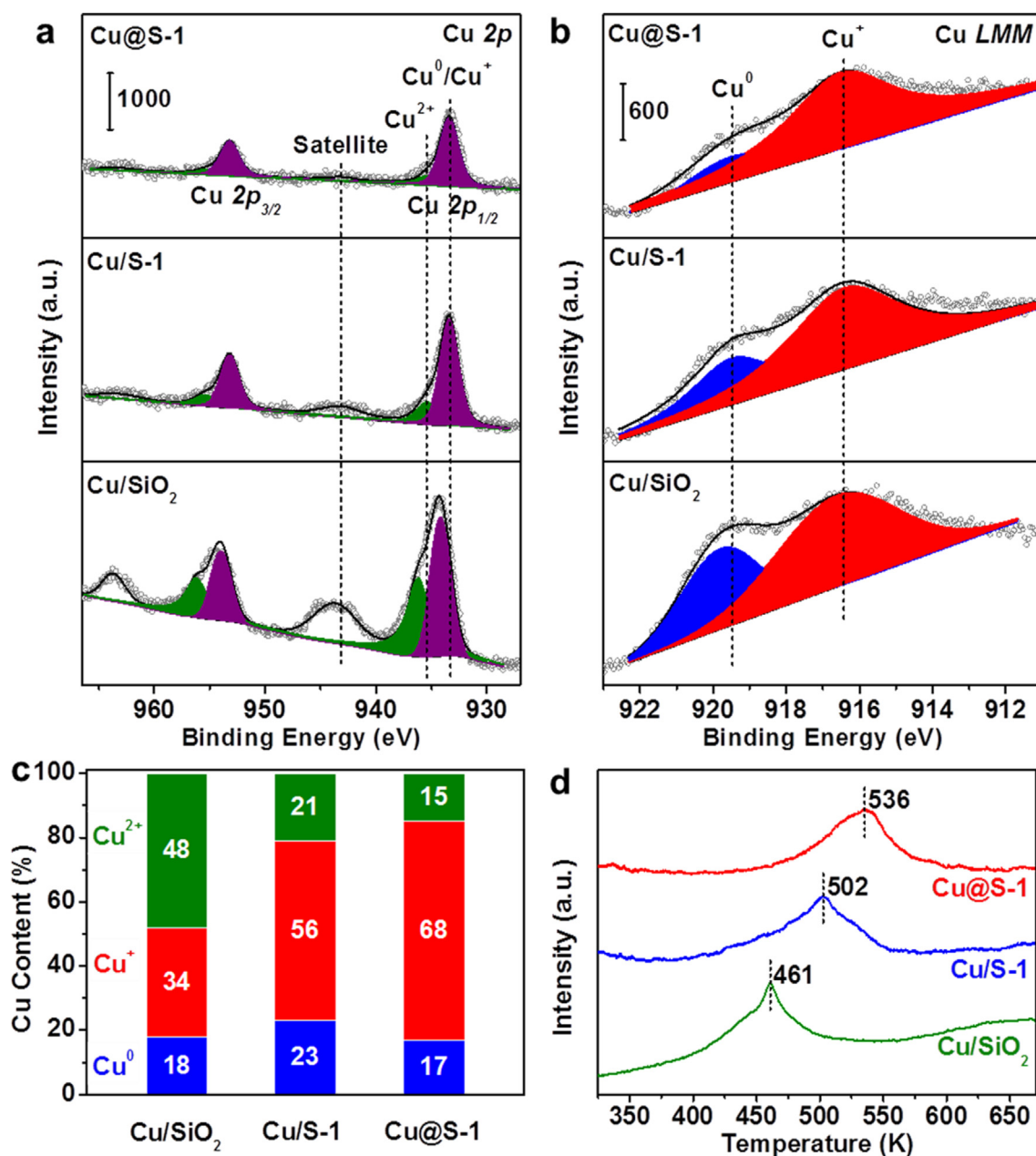


Fig. 5. (a) Cu 2p XPS spectra, (b) Cu LMM XAES spectra, (c) contents of different Cu species determined by Cu 2p XPS and Cu LMM XAES spectra, and (d) H₂-TPR profiles of Cu/SiO₂, Cu/S-1 and Cu@S-1.

Combined with the Cu 2p XPS results, the individual contents of Cu⁰, Cu⁺ and Cu²⁺ species in Cu/SiO₂, Cu/S-1 and Cu@S-1 catalysts can be estimated through normalizing the area of the corresponding featured peaks (Fig. 5c). The content of Cu²⁺ species in Cu/SiO₂ (48 %) is higher than Cu/S-1 (21 %) and Cu@S-1 (15 %). This might account for the more positively charged environment of Cu species in Cu/SiO₂ than that in Cu/S-1 and Cu@S-1, as indicated by FT-IR-CO and XPS. Notably, Cu@S-1 possesses a significant higher proportion of Cu⁺ species (68 %) compared with nonencapsulated counterparts (i.e. Cu/SiO₂ and Cu/S-1) obtained by wet impregnation. Such an enrichment of Cu⁺ species in Cu@S-1 was also observed for the encapsulated Cu nanoparticles in the zeolite TS-1 [33]. The prevailing Cu⁺ species might be stabilized by the silanol groups in the hydroxyl nests of the zeolite S-1, and such sites have been reported to be able to anchor Cu cations strongly in the zeolite confined environment [34].

Temperature-programmed reduction of hydrogen (H₂-TPR) measurements were also carried out on Cu/SiO₂, Cu/S-1 and Cu@S-1 catalysts. As shown in Fig. 5d, the reduction temperature of Cu species in Cu@S-1 (536 K) are higher than that of Cu/SiO₂ (461 K) and Cu/S-1 (502 K). Li and co-workers also reported that the reduction peaks of encapsulated Pd nanoparticles in Pd@S-1 were observed in the higher temperature range (443–523 K) than those of nonencapsulated Pd species in Pd/S-1 (423–503 K) [50]. Therefore, Cu species in Cu@S-1 are more difficult to be reduced than the counterparts in Cu/SiO₂ and Cu/S-1 during H₂ treatment. This further confirms the existence of an enhance interaction between the encapsulated Cu species and the silanol groups from S-1, which might stabilize Cu⁺ species and thus result into the predominant amount of Cu⁺ species in Cu@S-1.

The related contents of different Cu species in the spent Cu/SiO₂, Cu/S-1 and Cu@S-1 catalysts have also been characterized and

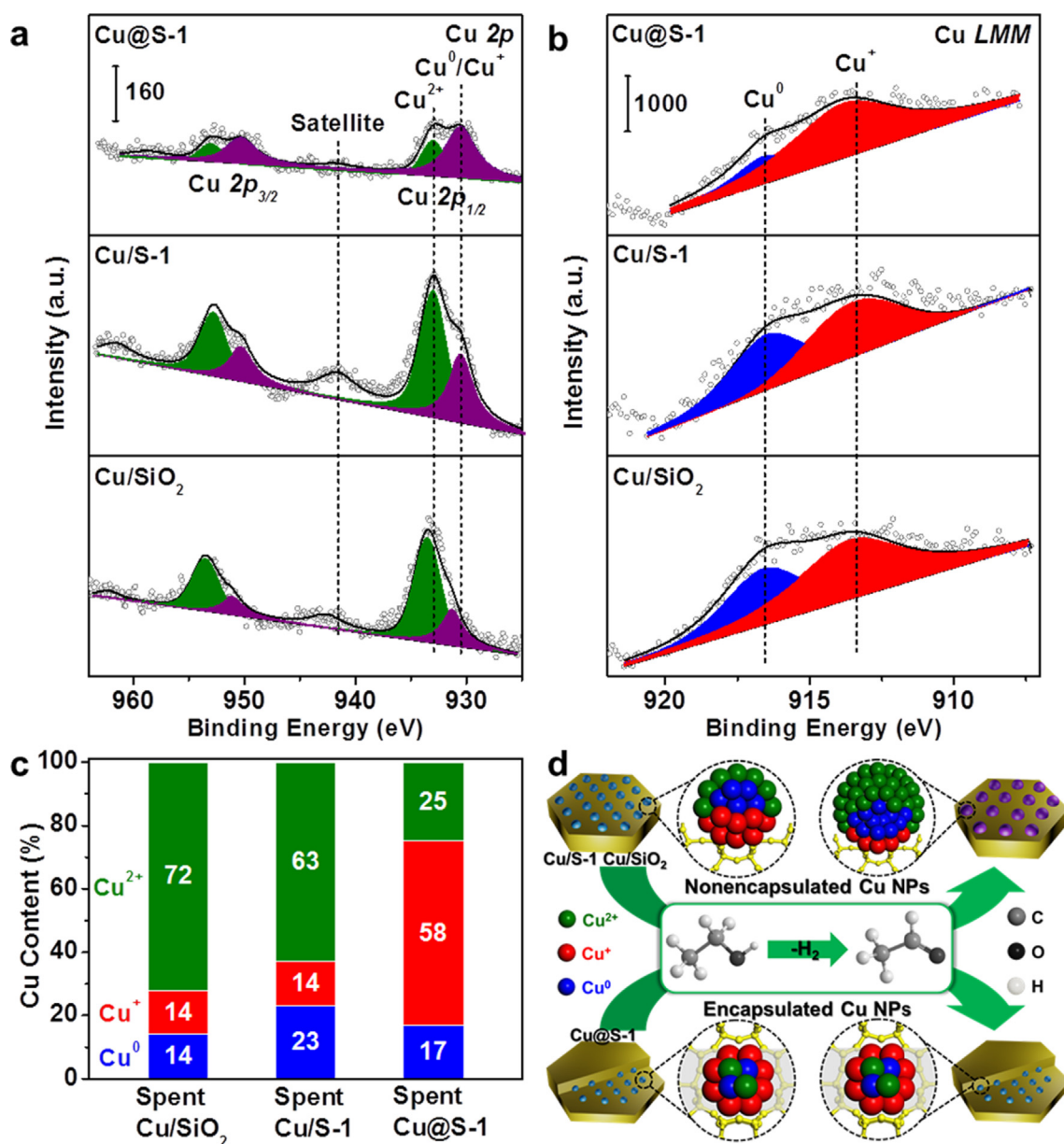


Fig. 6. (a) Cu 2p XPS spectra, (b) Cu LMM XAES spectra, (c) contents of different Cu species determined by Cu 2p XPS and Cu LMM XAES spectra of the spent Cu/SiO₂, Cu/S-1 and Cu@S-1 after a TOS of 16 h, 18 h, 110 h, respectively; (d) Schematic illustration of the structure of zeolite-encapsulated Cu nanoparticles in the Cu@S-1 in the dehydrogenation of ethanol to acetaldehyde.

quantified by the combination of Cu 2p XPS and Cu LMM XAES (Fig. 6a–c). Upon catalysis, Cu/SiO₂ shows a significant decrease in Cu⁺ species from 34 % to 14 % and an obvious increase in Cu²⁺ species from 48 % to 72 %, together with a slight decrease in metallic Cu⁰ species from 18 % to 14 %, after a TOS of 16 h. According to previous reports, Cu⁺ species have been experimentally and theoretically demonstrated to be the active sites of Cu-based catalyst in the dehydrogenation of ethanol to acetaldehyde, which could efficiently catalyze the cleavage of C–H and O–H bonds of ethanol and accelerate the adsorption and activation of ethanol [51,52]. Therefore, the decreased proportion of Cu⁺ species might negatively affect the activity of Cu/SiO₂ in ethanol dehydrogenation. In addition, the significant loss in porosity and an obvious increase in the average Cu particle size from ~49 to ~72 nm are observed for the spent Cu/SiO₂ by N₂ physisorption and N₂O chemisorption (Table 1), respectively. Those imply that the inferior performance and quick deactivation of Cu/SiO₂ are most likely caused by the combination of the severe carbon deposition, sintering of Cu nanoparticles and the instability of the catalytically active Cu⁺ sites. For Cu/S-1, a major decrease in Cu⁺ species from 56 % to 14 % and a simultaneous increase in Cu²⁺ species from 21 % to 63 % are observed after a TOS of 18 h. As limited leaching of Cu species has been indicated for the spent Cu-based catalysts (Table 1), the increase in Cu²⁺ species is majorly at the expense of the active Cu⁺ species in Cu/S-1 during catalysis. Besides, an increase of the average Cu particle size from 2.8 to 3.7 nm is detected for the spent Cu/S-1 by N₂O chemisorption (Table 1) and STEM images (Fig. S6). Therefore, the sintering of Cu nanoparticles and the decrease in the amount of the active Cu⁺ sites are still the inevitable issues, resulting into a gradual catalyst deactivation of Cu/S-1. Notably, Cu@S-1 retains the largest amount of the Cu⁺ species up to 58 % even after a TOS of 110 h, compared to Cu/S-1 and Cu/SiO₂. Note that such a Cu⁺ amount of 58 % in the spent Cu@S-1 is still slightly higher than that in the fresh Cu/S-1 (56 %), thus enabling the sustained superior catalytic performance. Besides, no apparent increase in the average Cu particle size is shown for the spent Cu@S-1 even after a TOS of 110 h. As such, the active Cu⁺ species can be efficiently preserved and the Cu sintering can be prevented for the encapsulated Cu@S-1 during ethanol dehydrogenation.

In view of all the results above, the superior catalytic performance of Cu@S-1 in the selective dehydrogenation of ethanol to acetaldehyde, in terms of activity, selectivity and stability, associates with not only the successful encapsulation of well-dispersed Cu nanoparticles but also the efficient preservation of the intrinsically active Cu⁺ species in the confined zeolite microenvironment. Even taking into account of the inevitable oxidation by the exposure to air during the sample storage and transportation, Cu@S-1 is shown to be the least affected catalyst with the highest proportion of Cu⁺ species and the lowest proportion of Cu²⁺ species among the three Cu-based catalysts. Zeolite-tailored encapsulation could not only restrict the movements of metal nanoparticles, leading to an enhanced stability by preventing metal sintering during catalysis, but also efficiently maintain the predominant Cu⁺ species stabilized by the spatially confined environment of the zeolite, resulting into the sustained superior performance even after a long-term continuous durability test (Fig. 6d).

4. Conclusions

In this contribution, well-dispersed Cu nanoparticles (~1.8 nm) encapsulated inside the zeolite S-1 have been successfully developed by an in-situ encapsulation strategy. The catalytic performances of Cu-based catalysts in the selective dehydrogenation of ethanol into acetaldehyde are found to be highly dependent on the nature of Cu encapsulation and the proportion of the intrinsi-

cally active Cu⁺ species. Zeolite-encapsulated Cu nanoparticles have afforded great benefits to catalytic performance, in terms of activity, selectivity, and stability, compared to the nonencapsulated Cu-based analogues prepared by wet impregnation. The as-prepared Cu@S-1 catalyst shows a sustained, superior performance of ethanol dehydrogenation, with an ethanol conversion of ~98 % and an acetaldehyde yield of ~90 % even after a TOS of 110 h at 523 K. While the Cu/SiO₂ and Cu/S-1 benchmark catalysts encounter with a quick/gradual deactivation within a TOS of 20 h. The combining multiple characterization techniques reveal the positive impacts of zeolite-tailored encapsulation on improving the catalyst performance, in particular catalyst stability. The successful encapsulation of Cu species within the zeolite S-1, could not only restrict the movements of metal nanoparticles and inhibit metal sintering during catalysis, but also achieve a significant proportion (up to ~70 %) of the active but instable Cu⁺ species, stabilized inside the zeolite due to an induced interaction between the encapsulated Cu nanoparticles and the spatially confined environment, which both result into an enhanced, sustained performance due to the efficient maximization and preservation of the intrinsic active sites during the long-term stability tests. This work could provide an efficient approach of zeolite-tailored encapsulation for the rational design and development of the highly-dispersed and well-defined Cu-based catalysts, for attaining an enhanced catalytic performance in ethanol dehydrogenation and other biomass-related conversion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The National Key R&D Program of China (2018YFB1501602), National Natural Science Foundation of China (22078316), National Key R&D Program of China (2021YFA1501203), and CNPC Innovation Found (2021DQ02-0702) are acknowledged for financial support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcat.2022.07.014>.

References

- [1] R.A. Sheldon, I.W.C.E. Arends, G.-J. Ten Brink, A. Dijkman, Green, Catalytic Oxidations of Alcohols, *Acc. Chem. Res.* 35 (9) (2002) 774–781.
- [2] L. Wu, T. Moteki, A.A. Gokhale, D.W. Flaherty, F.D. Toste, Production of Fuels and Chemicals from Biomass: Condensation Reactions and Beyond, *Chem* 1 (2016) 32–58.
- [3] C. Angelici, B.M. Weckhuysen, P.C.A. Bruijninx, Chemocatalytic Conversion of Ethanol into Butadiene and Other Bulk Chemicals, *ChemSusChem* 6 (9) (2013) 1595–1614.
- [4] X. Wu, G. Fang, Y. Tong, D. Jiang, Z. Liang, W. Leng, L. Liu, P. Tu, H. Wang, J. Ni, Catalytic Upgrading of Ethanol to *n*-Butanol: Progress in Catalyst Development, *ChemSusChem* 11 (2018) 71–85.
- [5] J. Pang, M. Yin, P. Wu, X. Li, H. Li, M. Zheng, T. Zhang, Advances in Catalytic Dehydrogenation of Ethanol to Acetaldehyde, *Green Chem.* 23 (20) (2021) 7902–7916.
- [6] R. Gérardy, D.P. Debecker, J. Estager, P. Luis, J.-C. Monbaliu, Continuous Flow Upgrading of Selected C₂–C₆ Platform Chemicals Derived from Biomass, *Chem. Rev.* 120 (15) (2020) 7219–7347.
- [7] G.W. Huber, S. Iborra, A. Corma, Synthesis of Transportation Fuels from Biomass: Chemistry, Catalysts, and Engineering, *Chem. Rev.* 106 (2006) 4044–4098.
- [8] D. Shindell, C.J. Smith, Climate and Air-Quality Benefits of A Realistic Phase-out of Fossil Fuels, *Nature* 573 (7774) (2019) 408–411.

- [9] C.O. Tuck, E. Pérez, I.T. Horváth, R.A. Sheldon, M. Poliakoff, Valorization of Biomass: Deriving More Value from Waste, *Science* 337 (2012) 695–699.
- [10] Y. Huang, B. Wang, H. Yuan, Y. Sun, D. Yang, X. Cui, F. Shi, The Catalytic Dehydrogenation of Ethanol by Heterogeneous Catalysts, *Catal. Sci. Technol.* 11 (5) (2021) 1652–1664.
- [11] E.V. Makshina, M. Dusselier, W. Janssens, J. Degrevé, P.A. Jacobs, B.F. Sels, Review of Old Chemistry and New Catalytic Advances in the On-Purpose Synthesis of Butadiene, *Chem. Soc. Rev.* 43 (22) (2014) 7917–7953.
- [12] J.M. Restrepo-Flórez, C.T. Maravelias, Advanced Fuels from Ethanol - A Superstructure Optimization Approach, *Energ. Environ. Sci.* 14 (1) (2021) 493–506.
- [13] M. Ni, D.Y.C. Leung, M.K.H. Leung, A Review on Reforming Bio-Ethanol for Hydrogen Production, *Int. J. Hydrogen Energ.* 32 (15) (2007) 3238–3247.
- [14] H. Zhang, M.Y.S. Ibrahim, D.W. Flaherty, Aldol Condensation among Acetaldehyde and Ethanol Reactants on TiO₂: Experimental Evidence for the Kinetically Relevant Nucleophilic Attack of Enolates, *J. Catal.* 361 (2018) 290–302.
- [15] N.M. Eagan, M.D. Kumbhalkar, J.S. Buchanan, J.A. Dumesic, G.W. Huber, Chemistries and Processes for the Conversion of Ethanol into Middle-Distillate Fuels, *Nat. Rev. Chem.* 3 (4) (2019) 223–249.
- [16] M. Behrens, F. Studt, I. Kasatkin, S. Kühl, M. Hävecker, F. Abild-Pedersen, S. Zander, F. Girgsdies, P. Kurr, B.-L. Knier, M. Tovar, R.W. Fischer, J.K. Nørskov, R. Schlögl, The Active Site of Methanol Synthesis over Cu/ZnO/Al₂O₃ Industrial Catalysts, *Science* 336 (6083) (2012) 893–897.
- [17] A. Chen, X. Yu, Y. Zhou, S. Miao, Y. Li, S. Kuld, J. Liu, T. Aoki, S. Hong, M.F. Camellone, S. Fabris, J. Ning, C. Jin, C. Yang, A. Nefedov, C. Wöll, Y. Wang, W. Shen, Structure of the Catalytically Active Copper-Ceria Interfacial Perimeter, *Nat. Catal.* 2 (4) (2019) 334–341.
- [18] M.B. Gawande, A. Goswami, F.-X. Felpin, T. Asefa, X. Huang, R. Silva, X. Zou, R. Zboril, R.S. Varma, Cu and Cu-Based Nanoparticles: Synthesis and Applications in Catalysis, *Chem. Rev.* 116 (2016) 3722–3811.
- [19] H. Yue, X. Ma, J. Gong, An Alternative Synthetic Approach for Efficient Catalytic Conversion of Syngas to Ethanol, *Acc. Chem. Res.* 47 (5) (2014) 1483–1492.
- [20] R.M.M. Santos, V. Briois, L. Martins, C.V. Santilli, Insights into the Preparation of Copper Catalysts Supported on Layered Double Hydroxide Derived Mixed Oxides for Ethanol Dehydrogenation, *ACS Appl. Mater. Inter.* 13 (22) (2021) 26001–26012.
- [21] C.H. Bartholomew, Mechanisms of Catalyst Deactivation, *Appl. Catal. A-Gen* 212 (1–2) (2001) 17–60.
- [22] H. Zhang, C. Canlas, A. Jeremy Kropf, J.W. Elam, J.A. Dumesic, C.L. Marshall, Enhancing the Stability of Copper Chromite Catalysts for the Selective Hydrogenation of Furfural with ALD Overcoating (II) - Comparison between TiO₂ and Al₂O₃ Overcoatings, *J. Catal.* 326 (2015) 172–181.
- [23] H. Zhang, Y.u. Lei, A.J. Kropf, G. Zhang, J.W. Elam, J.T. Miller, F. Sollberger, F. Ribeiro, M.C. Akatay, E.A. Stach, J.A. Dumesic, C.L. Marshall, Enhancing the Stability of Copper Chromite Catalysts for the Selective Hydrogenation of Furfural Using ALD Overcoating, *J. Catal.* 317 (2014) 284–292.
- [24] J. Pang, M. Zheng, C. Wang, X. Yang, H. Liu, X. Liu, J. Sun, Y. Wang, T. Zhang, Hierarchical Echinus-Like Cu-MFI Catalysts for Ethanol Dehydrogenation, *ACS Catal.* 10 (22) (2020) 13624–13629.
- [25] W.H. Cassinelli, L. Martins, A.R. Passos, S.H. Pulcinelli, A. Rochet, V. Briois, C.V. Santilli, Correlation between Structural and Catalytic Properties of Copper Supported on Porous Alumina for the Ethanol Dehydrogenation Reaction, *ChemCatChem* 7 (11) (2015) 1668–1677.
- [26] J. Sun, Y. Wang, Recent Advances in Catalytic Conversion of Ethanol to Chemicals, *ACS Catal.* 4 (4) (2014) 1078–1090.
- [27] H. Xin, X. Li, Y. Fang, X. Yi, W. Hu, Y. Chu, F. Zhang, A. Zheng, H. Zhang, X. Li, Catalytic Dehydration of Ethanol over Post-Treated ZSM-5 Zeolites, *J. Catal.* 312 (2014) 204–215.
- [28] J.T. Kozlowski, R.J. Davis, Sodium Modification of Zirconia Catalysts for Ethanol Coupling to 1-Butanol, *J. Energy Chem.* 22 (1) (2013) 58–64.
- [29] W. Luo, W. Cao, P.C.A. Bruijninx, L.u. Lin, A. Wang, T. Zhang, Zeolite-Supported Metal Catalysts for Selective Hydrodeoxygenation of Biomass-Derived Platform Molecules, *Green Chem.* 21 (14) (2019) 3744–3768.
- [30] J. Yang, Y. He, J. He, Y. Liu, H. Geng, S. Chen, L.u. Lin, M. Liu, T. Chen, Q. Jiang, B. M. Weckhuysen, W. Luo, Z. Wu, Enhanced Catalytic Performance through in Situ Encapsulation of Ultrafine Ru Clusters within a High-Aluminum Zeolite, *ACS Catal.* 12 (3) (2022) 1847–1856.
- [31] H.J. Cho, D. Kim, J. Li, D. Su, B. Xu, Zeolite-Encapsulated Pt Nanoparticles for Tandem Catalysis, *J. Am. Chem. Soc.* 140 (41) (2018) 13514–13520.
- [32] M. Limlamthong, A.C.K. Yip, Recent Advances in Zeolite-Encapsulated Metal Catalysts: A Suitable Catalyst Design for Catalytic Biomass Conversion, *Bioresour. Technol.* 297 (2020) 122488.
- [33] P. Cao, L.u. Lin, H. Qi, R. Chen, Z. Wu, N. Li, T. Zhang, W. Luo, Zeolite-Encapsulated Cu Nanoparticles for the Selective Hydrogenation of Furfural to Furfuryl Alcohol, *ACS Catal.* 11 (16) (2021) 10246–10256.
- [34] D. Yu, W. Dai, G. Wu, N. Guan, L. Li, Stabilizing Copper Species Using Zeolite for Ethanol Catalytic Dehydrogenation to Acetaldehyde, *Chin. J. Catal.* 40 (9) (2019) 1375–1384.
- [35] G.C. Chinchin, C.M. Hay, H.D. Vandervell, K.C. Waugh, The Measurement of Copper Surface-Areas by Reactive Frontal Chromatography, *J. Catal.* 103 (1987) 79–86.
- [36] M. Zaarour, J. Cazemier, J. Ruiz-Martínez, Recent Developments in the Control of Selectivity in Hydrogenation Reactions by Confined Metal Functionalities, *Catal. Sci. Technol.* 10 (24) (2020) 8140–8172.
- [37] H. Zhang, H.-R. Tan, S. Jaenicke, G.-K. Chuah, Highly Efficient and Robust Cu Catalyst for Non-Oxidative Dehydrogenation of Ethanol to Acetaldehyde and Hydrogen, *J. Catal.* 389 (2020) 19–28.
- [38] J. He, L.u. Lin, M. Liu, C. Miao, Z. Wu, R. Chen, S. Chen, T. Chen, Y. Su, T. Zhang, W. Luo, A Durable Ni/La-Y Catalyst for Efficient Hydrogenation of γ -Valerolactone into Pentanoic Biofuels, *J. Energy Chem.* 70 (2022) 347–355.
- [39] T. Hou, S. Zhang, Y. Chen, D. Wang, W. Cai, Hydrogen Production from Ethanol Reforming: Catalysts and Reaction Mechanism, *Renewable Sustainable Energy Rev.* 44 (2015) 132–148.
- [40] F. Giordano, P.N.R. Venneström, L.F. Lundegaard, F.N. Stappen, S. Mossin, P. Beato, S. Bordiga, C. Lamberti, Characterization of Cu-exchanged SSZ-13: A Comparative FTIR, UV-Vis, and EPR Study with Cu-ZSM-5 and Cu- β with Similar Si/Al and Cu/Al Ratios, *Dalton Trans.* 42 (2013) 12741–12761.
- [41] G.T. Palomino, S. Bordiga, A. Zecchina, G.L. Marra, C. Lamberti, XRD, XAS, and IR Characterization of Copper-Exchanged Y Zeolite, *J. Phys. Chem. B* 104 (36) (2000) 8641–8651.
- [42] A.M. Abdel-Mageed, B. Rungtaweeworant, M. Parlinska-Wojtan, X. Pei, O.M. Yaghi, R.J. Behm, Highly Active and Stable Single-Atom Cu Catalysts Supported by a Metal-Organic Framework, *J. Am. Chem. Soc.* 141 (13) (2019) 5201–5210.
- [43] J. Szanyi, J.H. Kwak, H. Zhu, C.H.F. Peden, Characterization of Cu-SSZ-13 NH₃ SCR catalysts: An In Situ FTIR Study, *Phys. Chem. Chem. Phys.* 15 (7) (2013) 2368–2380.
- [44] C. Lamberti, S. Bordiga, A. Zecchina, M. Salvalaggio, F. Geobaldo, C. Otero Areán, XANES, EXAFS and FTIR Characterization of Copper-Exchanged Mordenite, *J. Chem. Soc., Faraday Trans. 94* (1998) 1519–1525.
- [45] X. Shi, Y. Lin, L. Huang, Z. Sun, Y. Yang, X. Zhou, E. Vovk, X. Liu, X. Huang, M. Sun, S. Wei, J. Lu, Copper Catalysts in Semihydrogenation of Acetylene: From Single Atoms to Nanoparticles, *ACS Catal.* 10 (2020) 3495–3504.
- [46] X. Qiu, M. Miyauchi, K. Sunada, M. Minoshima, M. Liu, Y. Lu, D. Li, Y. Shimodaira, Y. Hosogi, Y. Kuroda, K. Hashimoto, Hybrid Cu₂O/TiO₂ Nanocomposites as Risk-Reduction Materials in Indoor Environments, *ACS Nano* 6 (2) (2012) 1609–1618.
- [47] X. Wang, S. Qiu, J. Feng, Y. Tong, F. Zhou, Q. Li, L.i. Song, S. Chen, K.-H. Wu, P. Su, S. Ye, F. Hou, S.X. Dou, H.K. Liu, G.Q. (Max) Lu, C. Sun, J. Liu, J.i. Liang, Confined Fe-Cu Clusters as Sub-Nanometer Reactors for Efficiently Regulating the Electrochemical Nitrogen Reduction Reaction, *Adv. Mater.* 32 (40) (2020) 2004382.
- [48] J.P. Espinós, J. Morales, A. Barranco, A. Caballero, J.P. Holgado, A.R. González-Elipé, Interface Effects for Cu, CuO, and Cu₂O Deposited on SiO₂ and ZrO₂. XPS Determination of the Valence State of Copper in Cu/SiO₂ and Cu/ZrO₂ Catalysts, *J. Phys. Chem. B* 106 (27) (2002) 6921–6929.
- [49] A. Yin, X. Guo, W.-L. Dai, K. Fan, The Nature of Active Copper Species in Cu-HMS Catalyst for Hydrogenation of Dimethyl Oxalate to Ethylene Glycol: New Insights on the Synergetic Effect between Cu⁰ and Cu⁺, *J. Phys. Chem. C* 113 (2009) 11003–11013.
- [50] Y. Chai, S. Liu, Z.-J. Zhao, J. Gong, W. Dai, G. Wu, N. Guan, L. Li, Selectivity Modulation of Encapsulated Palladium Nanoparticles by Zeolite Microenvironment for Biomass Catalytic Upgrading, *ACS Catal.* 8 (9) (2018) 8578–8589.
- [51] S. Hanukovich, A. Dang, P. Christoffer, Influence of Metal Oxide Support Acid Sites on Cu-Catalyzed Nonoxidative Dehydrogenation of Ethanol to Acetaldehyde, *ACS Catal.* 9 (4) (2019) 3537–3550.
- [52] X. He, Y. Wang, X. Zhang, M. Dong, G. Wang, B. Zhang, Y. Niu, S. Yao, X. He, H. Liu, Controllable in Situ Surface Restructuring of Cu Catalysts and Remarkable Enhancement of Their Catalytic Activity, *ACS Catal.* 9 (3) (2019) 2213–2221.